COMPUTER NETWORKS

Complete Notes

First Year → Final Year | Interview Ready

Covers OSI Model • TCP/IP • Subnetting • Routing • Security
Wireless • DNS • HTTP/HTTPS • Network Devices & More

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20–30 Pages • Diagrams & Examples • Job Switch Ready

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CHAPTER 1

Introduction to Computer Networks

Fundamentals every CS student must know

A **Computer Network** is a collection of interconnected devices (computers, servers, printers, etc.) that can communicate and share resources. Networks are the backbone of the internet, enterprise infrastructure, and every modern application you use today.

Types of Networks

Type	Full Form	Range	Example
PAN	Personal Area Network	< 10 m	Bluetooth headphones
LAN	Local Area Network	10m–1km	Office, school campus
MAN	Metropolitan Area Network	1–100km	City-wide ISP
WAN	Wide Area Network	> 100km	Internet, MPLS links
CAN	Campus Area Network	1–5 km	University campus
SAN	Storage Area Network	Local	Data center storage

Network Topologies

Topology defines **how devices are physically or logically arranged** in a network.

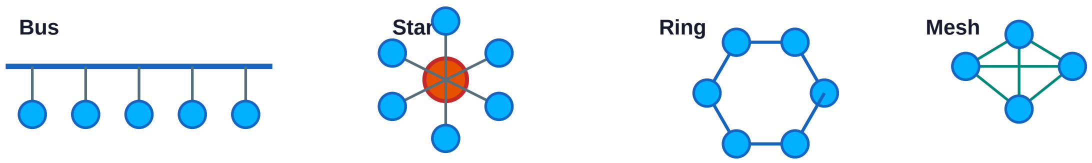


Fig 1.1 — Common Network Topologies

Topology	Advantage	Disadvantage	Use Case
Bus	Simple, low cost	Single point of failure	Old Ethernet
Star	Easy to manage	Hub/switch dependency	Modern LAN
Ring	Equal access	One failure breaks ring	Token Ring
Mesh	Highly fault tolerant	Expensive cabling	WAN / Internet
Tree	Scalable hierarchy	Root failure is critical	Enterprise LAN
Hybrid	Flexible, scalable	Complex design	Large networks

Transmission Media

Guided (Wired) Media
• Twisted Pair (UTP/STP) — Cat5e (1 Gbps), Cat6 (10 Gbps) — Used in LAN
• Coaxial Cable — 10Base2/10Base5 — TV, old Ethernet

• Optical Fiber — Single-mode (long distance), Multi-mode (short) — 100 Gbps+

Unguided (Wireless) Media
• Radio Waves — Wi-Fi (2.4/5/6 GHz), Bluetooth (2.4 GHz)
• Microwaves — Satellite, Point-to-point links
• Infrared — TV remotes, short range (~1m)

■ *Interview Tip: Single-mode fiber uses laser light and supports distances up to 100 km; multi-mode uses LEDs and covers up to 2 km. Critical for data center questions.*

CHAPTER 2

OSI Model & TCP/IP Model

The most important concept in Computer Networks

OSI Reference Model (7 Layers)

The **Open Systems Interconnection (OSI)** model is a conceptual framework for understanding how data travels across a network. Each layer has a specific responsibility and communicates only with the layers directly above and below it.

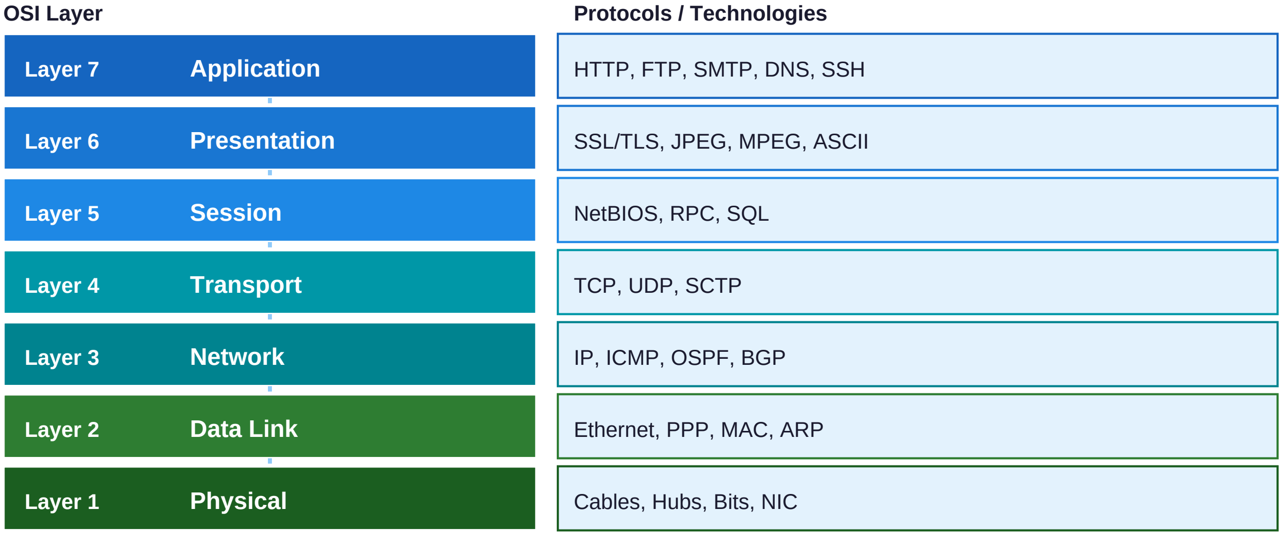


Fig 2.1 — OSI 7-Layer Model with Protocols

Layer	Name	Function	PDU	Device
7	Application	User interface, network services	Data	Proxy, FW
6	Presentation	Encryption, compression, translation	Data	—
5	Session	Session management, sync	Data	—
4	Transport	End-to-end delivery, error recovery	Segment	FW, LB
3	Network	Logical addressing, routing	Packet	Router
2	Data Link	Physical addressing (MAC), framing	Frame	Switch
1	Physical	Bit transmission over medium	Bit	Hub, NIC

TCP/IP Model vs OSI Model

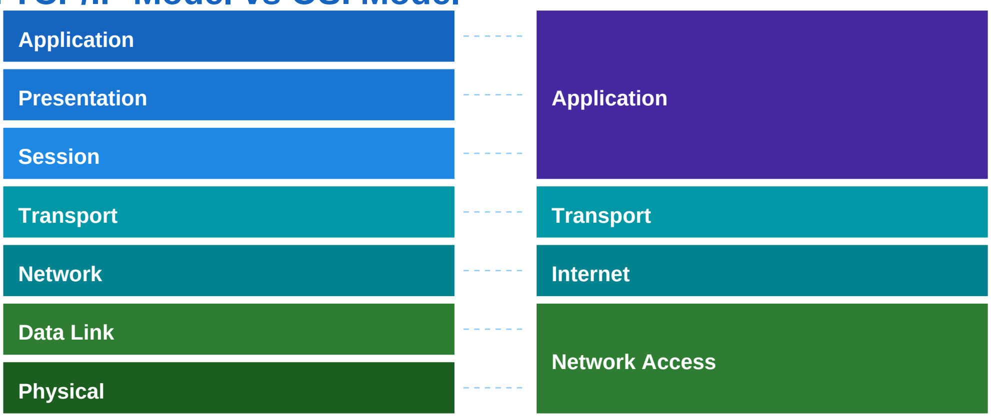


Fig 2.2 — OSI vs TCP/IP Layer Mapping

Encapsulation / De-encapsulation
• Sender adds headers at each layer going DOWN (Application → Physical)
• Receiver removes headers at each layer going UP (Physical → Application)
• Application Layer: Data → Transport: Segment → Network: Packet → DLL: Frame → Physical: Bits
• Mnemonic for OSI layers (top→bottom): "All People Seem To Need Data Processing"

■ *Most asked interview question: "What happens when you type www.google.com?" Answer involves DNS, TCP handshake, HTTP, and all OSI layers!*

CHAPTER 3

Physical Layer

Signals, Encoding & Multiplexing

Signals & Bandwidth

The Physical Layer is responsible for transmitting raw **bits** over a physical medium. It deals with voltage levels, timing, and physical connectors.

Concept	Definition	Formula/Value
Bandwidth	Max data rate of a channel	B = f_max - f_min (Hz)
Throughput	Actual data transferred	Throughput ≤ Bandwidth
Latency	Time for data to travel	Propagation + Transmission + Queue
Nyquist Limit	Max rate, noiseless channel	2 × B × log2(L) bps
Shannon Capacity	Max rate, noisy channel	B × log2(1 + SNR) bps
Attenuation	Signal weakening with distance	Increases with distance
Noise	Unwanted signal interference	Thermal, crosstalk, IMD

Encoding Schemes

- NRZ-L (Non-Return to Zero Level): High voltage = 1, Low = 0
- NRZ-I (NRZ Inverted): Transition at beginning of 1 bit
- Manchester: Transition at middle of each bit; Low→High = 1, High→Low = 0 (Ethernet)
- Differential Manchester: Always transitions at mid-bit; no transition at start = 1
- 4B/5B: Maps 4-bit groups to 5-bit codes to avoid long sequences of zeros
- AMI (Alternate Mark Inversion): 0 = no signal, 1 alternates +V and -V

Multiplexing

Type	Full Form	Mechanism	Used In
FDM	Frequency Division MUX	Different frequency bands	Cable TV, Radio
TDM	Time Division MUX	Time slots per user	T1/E1 lines
WDM	Wavelength Division MUX	Different light wavelengths	Optical Fiber
CDM	Code Division MUX	Unique code per user	3G/CDMA cellular
OFDM	Orthogonal FDM	Overlapping subcarriers	LTE/Wi-Fi

Switching Techniques

Circuit Switching: Dedicated path established before communication (e.g., traditional telephone). Resources are reserved throughout the call.

Packet Switching: Data divided into packets, each routed independently (e.g., Internet). No dedicated path — efficient but variable delay.

Message Switching: Entire message stored then forwarded (store-and-forward). Obsolete — used in older telegraph/email systems.

Feature	Circuit Switching	Packet Switching
Path	Dedicated	Dynamic (per packet)

Delay	Low (fixed)	Variable
Bandwidth	Wasted if idle	Efficient
Failure handling	Entire call drops	Packets rerouted
Example	PSTN telephone	Internet (IP)

CHAPTER 4

Data Link Layer

Framing, MAC, Error Control & Flow Control

MAC Addressing & Framing

The Data Link Layer provides **node-to-node delivery**. It encapsulates packets into **frames** and adds MAC (Media Access Control) addresses.

MAC Address Facts
• 48-bit (6-byte) hardware address — written as XX:XX:XX:XX:XX:XX (hex)
• First 3 bytes = OUI (Organizationally Unique Identifier) — identifies manufacturer
• Last 3 bytes = Device-specific identifier assigned by manufacturer
• FF:FF:FF:FF:FF:FF = Broadcast address (sent to all devices on LAN)
• MAC addresses operate at Layer 2; IP addresses at Layer 3

Error Detection & Correction

Method	Type	How It Works	Detects/Corrects
Parity Bit	Detection	Add bit to make 1s even/odd	Single-bit errors
CRC	Detection	Polynomial division remainder	Burst errors (Ethernet)
Checksum	Detection	Sum of data segments	Multiple errors (TCP/IP)
Hamming Code	Correction	Add redundant bits at power-of-2 pos	Single-bit + detect double
Reed-Solomon	Correction	Polynomial-based block code	Burst errors (CDs, QR)

Flow Control Protocols

Stop & Wait: Send 1 frame, wait for ACK. Efficiency = 1/(1 + 2a) where a = propagation delay/transmission time

Go-Back-N: Send up to N frames, but on error retransmit from error frame onward. Window size = 2^n - 1

Selective Repeat: Send up to N frames, retransmit only errored frames. Window size = 2^(n-1)

CSMA/CD & CSMA/CA

CSMA/CD (Carrier Sense Multiple Access / Collision Detection): Used in wired Ethernet. Listen before transmit; if collision detected, stop → jam signal → random backoff → retry. Minimum frame size = 64 bytes.

CSMA/CA (Collision Avoidance): Used in Wi-Fi (802.11). Cannot detect collision in wireless; instead uses RTS/CTS handshake and random backoff timers to avoid collisions before they happen.

■ *Key difference: CD = detect after collision (wired), CA = avoid before collision (wireless). Both use random exponential backoff after failure.*

CHAPTER 5

Network Layer

IP Addressing, Subnetting & Routing

IPv4 Addressing

IPv4 uses 32-bit addresses written in dotted-decimal notation (e.g., 192.168.1.100). The address has two parts: **Network ID** and **Host ID**.

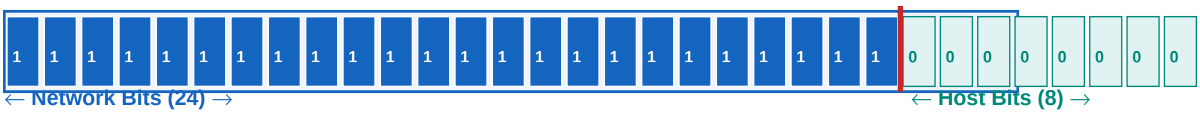
Class	Range	Default Mask	Networks	Hosts/Net	Use
A	1.0.0.0 – 126.x.x.x	/8 (255.0.0.0)	126	16,777,214	Large orgs
B	128.0.0.0 – 191.255.x.x	/16 (255.255.0.0)	16,384	65,534	Medium orgs
C	192.0.0.0 – 223.255.255.x	/24 (255.255.255.0)	2,097,152	254	Small nets
D	224.0.0.0 – 239.255.255.255	N/A	N/A	N/A	Multicast
E	240.0.0.0 – 255.255.255.255	N/A	N/A	N/A	Research

IPv4 Header Structure

Version (4b)	IHL (4b)	DSCP/ECN (8b)	Total Length (16b)
Identification (16b)		Flags (3b)	Fragment Offset (13b)
TTL (8b)		Protocol (8b)	Header Checksum (16b)
Source IP Address (32 bits)			
Destination IP Address (32 bits)			
Options (Variable)			Data / Payload →

Fig 5.1 — IPv4 Packet Header (20–60 bytes)

Subnetting & CIDR



Example: 192.168.1.0/24 → 256 addresses, 254 usable hosts

Fig 5.2 — /24 Subnet Bit Breakdown

Subnetting Quick Formula
• Number of subnets = 2^(borrowed bits)
• Number of hosts per subnet = 2^(host bits) - 2 (subtract network & broadcast)
• CIDR /24 → 256 addresses, 254 usable | /25 → 128 addr, 126 usable | /26 → 64 addr, 62 usable
• Private IP Ranges: 10.0.0.0/8 | 172.16.0.0/12 | 192.168.0.0/16
• Loopback: 127.0.0.1 | APIPA: 169.254.x.x | Broadcast: 255.255.255.255

IPv4 vs IPv6

Feature	IPv4	IPv6
Address Size	32 bits (4 bytes)	128 bits (16 bytes)

Notation	Dotted decimal	Colon-hex (8 groups)
Total Addresses	~4.3 billion	~3.4 × 10^38
Header Size	20–60 bytes	40 bytes (fixed)
Fragmentation	Routers & hosts	Only source host
Checksum	Yes (header)	No (handled by transport)
NAT Required	Yes (address exhaustion)	No (enough addresses)
Security	Optional (IPSec)	Mandatory (IPSec)
Config	Manual/DHCP	SLAAC/DHCPv6/Manual

■ *IPv6 addresses are written as 2001:0db8:85a3::8a2e:0370:7334 (:: replaces consecutive groups of zeros). Loopback = ::1*

CHAPTER 6

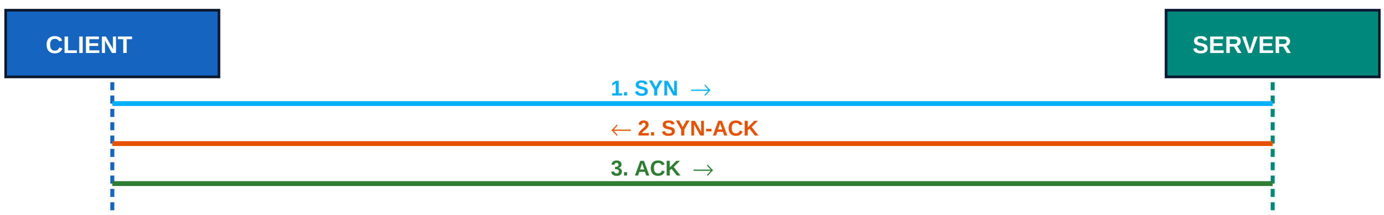
Transport Layer

TCP, UDP, Ports, Congestion Control

TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Guaranteed delivery	Best effort
Order	In-order delivery	No ordering
Flow Control	Yes (sliding window)	No
Error Control	Yes (retransmit)	Checksum only
Speed	Slower	Faster
Header Size	20–60 bytes	8 bytes
Use Cases	HTTP, FTP, SSH, SMTP	DNS, DHCP, VoIP, Video stream

TCP 3-Way Handshake



(Connection Established — 3-Way Handshake)

Fig 6.1 — TCP Connection Establishment (3-Way Handshake)

TCP Connection Termination (4-Way)

- 1. Client → FIN → Server (I want to close)
- 2. Server → ACK → Client (OK, noted)
- 3. Server → FIN → Client (I am also done)
- 4. Client → ACK → Server (Confirmed — connection closed)
- TIME_WAIT state lasts 2×MSL (Maximum Segment Lifetime) ≈ 2 minutes

Important Port Numbers

Port	Protocol	Service
20, 21	FTP	File Transfer (data/control)
22	SSH	Secure Shell
23	Telnet	Remote terminal (insecure)
25	SMTP	Email sending
53	DNS	Domain Name System (TCP+UDP)
67/68	DHCP	IP address assignment

80	HTTP	Web traffic
110	POP3	Email retrieval
143	IMAP	Email (keeps on server)
443	HTTPS	Secure web (TLS)
3306	MySQL	Database
3389	RDP	Remote Desktop
8080	HTTP-Alt	Dev web servers

■ TCP Congestion Control

- **Slow Start:** Begin with cwnd=1 MSS, double every RTT until ssthresh reached
- **Congestion Avoidance:** After ssthresh: increase cwnd by 1 MSS per RTT (linear)
- **Fast Retransmit:** After 3 duplicate ACKs: retransmit without waiting for timeout
- **Fast Recovery:** After fast retransmit: ssthresh = cwnd/2, cwnd = ssthresh + 3

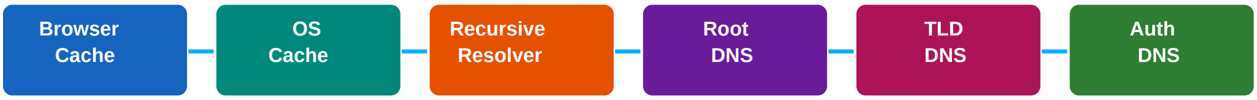
CHAPTER 7

Application Layer

DNS, HTTP/HTTPS, Email, DHCP & More

DNS (Domain Name System)

DNS translates human-readable domain names (e.g., **www.google.com**) into IP addresses. It is a **distributed hierarchical database**.



DNS Resolution Order: Browser Cache → OS → Resolver → Root → TLD → Authoritative

Fig 7.1 — DNS Resolution Flow

DNS Record Types

- A Record — Maps hostname → IPv4 address (e.g., example.com → 93.184.216.34)
- AAAA Record — Maps hostname → IPv6 address
- CNAME — Canonical Name (alias): www → example.com
- MX Record — Mail Exchange server for a domain
- NS Record — Nameserver for the domain
- PTR Record — Reverse DNS lookup (IP → domain)
- TXT Record — Arbitrary text (used for SPF, DKIM, verification)
- TTL — Time To Live: how long the record is cached

HTTP vs HTTPS

Feature	HTTP	HTTPS
Port	80	443
Security	None (plaintext)	TLS/SSL encrypted
Certificate	Not required	SSL/TLS cert required
Speed	Slightly faster	Slight overhead (negligible)
SEO	Penalized	Google preferred
Use	Internal/dev	All production sites

◆ HTTP Methods (REST verbs)

Method	Purpose	Idempotent	Safe
GET	Retrieve resource	Yes	Yes
POST	Create resource	No	No
PUT	Replace resource	Yes	No
PATCH	Partial update	No	No
DELETE	Delete resource	Yes	No

HEAD	GET without body	Yes	Yes
OPTIONS	List allowed methods	Yes	Yes

◆ HTTP Status Codes

Range	Category	Common Examples
1xx	Informational	100 Continue, 101 Switching Protocols
2xx	Success	200 OK, 201 Created, 204 No Content
3xx	Redirection	301 Moved Permanently, 302 Found, 304 Not Modified
4xx	Client Error	400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found
5xx	Server Error	500 Internal Server Error, 502 Bad Gateway, 503 Service Unavailable

■ Email Protocols

Protocol	Port	Role	Direction
SMTP	25 / 587	Send emails	Client → Server, Server → Server
POP3	110 / 995	Download emails (deletes from server)	Server → Client
IMAP	143 / 993	Access emails (keeps on server)	Server → Client (sync)

■ DHCP — Dynamic Host Configuration Protocol

DHCP automatically assigns IP addresses to devices on a network. Uses **UDP ports 67 (server) and 68 (client)**.

- DISCOVER — Client broadcasts "I need an IP address!" (src: 0.0.0.0, dst: 255.255.255.255)
- OFFER — Server offers an available IP address
- REQUEST — Client accepts the offer and requests that IP
- ACK — Server confirms the lease (DORA process)

CHAPTER 8

Network Devices

Hub, Switch, Router, Firewall & More

Core Network Devices

Device	OSI Layer	Function	Intelligence	Use Case
Hub	L1 Physical	Broadcast to all ports	None (dumb)	Old LAN (obsolete)
Switch	L2 Data Link	Forward by MAC address	MAC table	Modern LAN
Router	L3 Network	Forward by IP address	Routing table	LAN↔WAN, Internet
Bridge	L2 Data Link	Connect two LANs	MAC-based	Segment LAN
Gateway	L7 App	Protocol translation	Full stack	Protocol converter
Repeater	L1 Physical	Amplify signal	None	Extend cable distance
Modem	L1/L2	Modulate/demodulate	Minimal	DSL/Cable Internet
NIC	L1/L2	Physical network access	Hardware	Every networked device

Firewall & NAT

Firewall: A security system that monitors and controls incoming/outgoing network traffic based on predetermined security rules. Types: Packet Filtering, Stateful Inspection, Application-layer (Proxy), Next-Gen (NGFW).

NAT (Network Address Translation): Translates private IP addresses to public IPs before packets leave the network. Conserves IPv4 addresses. Types: SNAT (Static), DNAT (Dynamic), PAT/Overloading (many-to-one via ports).

VLAN (Virtual LAN)

- Logically segments a physical network without requiring separate hardware
- Configured on managed switches — devices in same VLAN communicate at L2
- Inter-VLAN routing requires a router or Layer 3 switch
- Tagged (802.1Q) frames carry VLAN ID in the header (trunk ports)
- Benefits: Security isolation, reduced broadcast domain, flexible topology

CHAPTER 9

Routing Protocols

Static, Dynamic, RIP, OSPF, BGP

Routing Basics

Routing is the process of **selecting paths** in a network to send data from source to destination. A router maintains a **routing table** to make forwarding decisions.

Feature	Static Routing	Dynamic Routing
Configuration	Manual by admin	Automatic (protocol learns)
Adaptability	No auto-adapt to changes	Adapts to topology changes
Overhead	None	Protocol messages consume bandwidth
Scalability	Poor for large nets	Scales well
Security	More secure (no protocol)	Protocol can be attacked
Use Case	Small/simple networks	Enterprise, ISP networks

Routing Algorithms

◆ Distance Vector (Bellman-Ford)

Each router knows distance to neighbors and shares its routing table periodically. Used by **RIP**. Problem: Slow convergence, count-to-infinity problem.

◆ Link State (Dijkstra's SPF)

Each router has complete topology map. Builds SPT using Dijkstra's algorithm. Used by **OSPF, IS-IS**. Fast convergence, scalable.

◆ Path Vector

Extension of DV; stores full path info. Used by **BGP** (internet routing). Prevents routing loops via AS path attribute.

Key Routing Protocols

Protocol	Type	Algorithm	Metric	AD	Use Case
RIP v1/v2	IGP / DV	Bellman-Ford	Hop count (max 15)	120	Small networks
OSPF	IGP / LS	Dijkstra SPF	Cost (bandwidth)	110	Enterprise LANs
EIGRP	IGP / Hybrid	DUAL	Composite metric	90	Cisco networks
BGP	EGP / PV	Path Vector	AS Path + attrs	20	Internet routing
IS-IS	IGP / LS	Dijkstra SPF	Cost	115	ISP backbone

■ *AD = Administrative Distance — lower is more trusted. Connected=0, Static=1, EIGRP=90, OSPF=110, RIP=120. If two protocols know same route, lower AD wins.*

CHAPTER 10

Network Security

Cryptography, Attacks, SSL/TLS & VPN

Cryptography Fundamentals

Type	Key Usage	Speed	Examples	Use Case
Symmetric	Same key encrypt/decrypt	Fast	AES, DES, 3DES, RC4	Bulk data encryption
Asymmetric	Public+Private key pair	Slow	RSA, ECC, Diffie-Hellman	Key exchange, digital signatures
Hash Function	No key (one-way)	Very Fast	MD5, SHA-1, SHA-256	Integrity verification
HMAC	Hash + secret key	Fast	HMAC-SHA256	Message authentication

SSL/TLS Handshake

- Step: ClientHello — Client sends supported TLS version, cipher suites, random number
- Step: ServerHello — Server chooses cipher suite, sends certificate + random number
- Step: Key Exchange — Client verifies certificate, generates pre-master secret, encrypts with server public key
- Step: Session Keys — Both derive symmetric session keys from pre-master secret
- Step: Finished — Both send encrypted "Finished" message; secure channel established

Common Network Attacks & Defenses

Attack	Type	Description	Defense
DDoS	Availability	Flood target with traffic from botnets	Rate limiting, CDN, scrubbing centers
Man-in-Middle	Interception	Intercept communication between two parties	TLS/HTTPS, certificate pinning
ARP Spoofing	L2 Attack	Fake ARP replies to redirect traffic	Dynamic ARP Inspection (DAI)
DNS Spoofing	DNS Attack	Corrupt DNS cache with fake records	DNSSEC, DoH, DoT
SQL Injection	App Layer	Inject malicious SQL via input fields	Parameterized queries, WAF
Port Scanning	Recon	Probe open ports (Nmap)	Firewall rules, IDS/IPS
Phishing	Social Eng	Fake emails/sites to steal credentials	User training, email filters
SYN Flood	DoS	Send many SYN without ACK (half-open)	SYN cookies, rate limiting

VPN (Virtual Private Network)

A VPN creates an **encrypted tunnel** over a public network, allowing secure remote access. Types: Site-to-Site VPN (connects two networks), Remote Access VPN (user connects to org network).

VPN Protocols
• IPSec — Uses AH + ESP; operates at Layer 3; common for site-to-site
• SSL/TLS VPN — Browser-based; uses HTTPS port 443; easy firewall traversal
• OpenVPN — Open-source; uses SSL/TLS; highly configurable
• WireGuard — Modern, lightweight, fast; uses UDP; gaining popularity
• L2TP/IPSec — L2TP for tunneling + IPSec for encryption; common on mobile

CHAPTER 11

Wireless Networks

Wi-Fi, Bluetooth, 5G & Security

Wi-Fi Standards (IEEE 802.11)

Standard	Name	Freq Band	Max Speed	Range	Year
802.11b	—	2.4 GHz	11 Mbps	35m	1999
802.11a	—	5 GHz	54 Mbps	35m	1999
802.11g	—	2.4 GHz	54 Mbps	38m	2003
802.11n	Wi-Fi 4	Dual	600 Mbps	70m	2009
802.11ac	Wi-Fi 5	5 GHz	3.5 Gbps	35m	2013
802.11ax	Wi-Fi 6/6E	2.4/5/6	9.6 Gbps	30m	2019
802.11be	Wi-Fi 7	Multi	46 Gbps	—	2024

Wi-Fi Security Protocols

Protocol	Year	Encryption	Vulnerability	Status
WEP	1997	RC4 (40/104 bit)	Easily cracked in minutes	Obsolete
WPA	2003	TKIP/RC4	TKIP weak, MITM possible	Deprecated
WPA2	2004	AES-CCMP	KRACK attack (patched)	Common
WPA3	2018	AES-GCMP-256	Side-channel (minor)	Current best

Cellular Networks Evolution

Generation	Year	Peak Speed	Technology	Use Case
1G	1980s	~2.4 Kbps	AMPS (Analog)	Voice only
2G	1990s	~144 Kbps	GSM, CDMA	Voice + SMS
3G	2000s	~7.2 Mbps	WCDMA, HSPA	Mobile internet
4G LTE	2009	~100 Mbps	OFDMA, MIMO	HD video, apps
5G	2019	~20 Gbps	mmWave, Massive MIMO	IoT, AR/VR, autonomous

■ *5G uses three spectrum bands: Low-band (wide coverage), Mid-band (balanced), High-band/mmWave (ultra-fast, short range). Core uses NFV and SDN.*

CHAPTER 12

Advanced Topics & Interview Prep

SDN, Cloud Networking, CDN & Top Interview Q&A;

■ SDN (Software-Defined Networking)

SDN separates the **control plane** (decisions) from the **data plane** (forwarding). A centralized software **controller** programs network devices via APIs (e.g., OpenFlow). Enables programmability and automation.

SDN vs Traditional Networking

- Traditional: Control + Data plane tightly coupled on each device
- SDN: Control plane centralized in software controller; data plane in hardware
- Benefits: Centralized management, dynamic configuration, vendor-agnostic
- Protocols: OpenFlow, NETCONF/YANG, REST APIs
- Used by: Google B4, Facebook Express Backbone, Hyperscalers

■ Cloud & Overlay Networking

Concept	Description	Examples
VPC	Virtual Private Cloud — isolated network in cloud	AWS VPC, GCP VPC, Azure VNet
Overlay Network	Virtual network on top of physical network	VXLAN, GRE, NVGRE
VXLAN	Virtual Extensible LAN — 24-bit VNI (16M segments)	Kubernetes, VMware NSX
Load Balancer	Distributes traffic across servers	AWS ALB/NLB, HAProxy, Nginx
CDN	Content Delivery Network — edge caching	Cloudflare, Akamai, AWS CloudFront
Service Mesh	Microservices L7 networking + security	Istio, Linkerd, Consul Connect

■ Top Interview Questions & Answers

Q1. What happens when you type "www.google.com" in a browser?

→ DNS resolves the domain → TCP 3-way handshake → TLS handshake (HTTPS) → HTTP GET request → Server responds with HTML → Browser renders page. Involves all OSI layers, ARP, routing, and application protocols.

Q2. Difference between TCP and UDP?

→ TCP: Connection-oriented, reliable, ordered, flow-controlled (HTTP, SSH, FTP). UDP: Connectionless, faster, no guarantee (DNS, VoIP, video streaming, gaming).

Q3. What is the difference between a router and a switch?

→ Switch operates at L2, uses MAC addresses, connects devices within a LAN. Router operates at L3, uses IP addresses, connects different networks (LAN↔WAN).

Q4. What is ARP and how does it work?

→ ARP (Address Resolution Protocol) maps IP → MAC address. Broadcasts "Who has IP x.x.x.x?" Target device replies with its MAC. Result cached in ARP table. Vulnerable to ARP spoofing attacks.

Q5. Explain subnetting. What is /28?

→ /28 = 28 network bits, 4 host bits → 2^4 = 16 addresses, 14 usable hosts. Subnet mask = 255.255.255.240. Subnetting divides a network into smaller segments to reduce broadcast domains and improve security.

Q6. What is NAT and why is it used?

→ NAT translates private IP addresses to a public IP. Solves IPv4 address exhaustion. Types: SNAT (static 1-to-1), DNAT (dynamic pool), PAT/overload (many-to-one via port numbers).

Q7. Difference between HTTP/1.1, HTTP/2, and HTTP/3?

→ HTTP/1.1: Text-based, one request/connection (keep-alive added). HTTP/2: Binary framing, multiplexing, header compression (HPACK), server push. HTTP/3: Uses QUIC (UDP-based), faster handshake, connection migration support.

Q8. What is a CDN and how does it work?

→ CDN (Content Delivery Network) caches content on edge servers near users. Reduces latency, offloads origin server, improves availability. DNS-based routing sends user to nearest PoP (Point of Presence).

Q9. What is OSPF and how does it work?

→ OSPF (Open Shortest Path First) is a Link-State IGP using Dijkstra's SPF algorithm. Each router floods LSAs to build identical LSDB. Calculates shortest path tree. Uses areas (Area 0 = backbone) for scalability. AD = 110, metric = cost.

Q10. How does TLS/SSL work?

→ Asymmetric crypto for key exchange → Symmetric crypto for session data. Client verifies server certificate (CA-signed). Derives session keys via key exchange. Modern TLS 1.3 eliminates RSA key exchange, only uses ECDHE for forward secrecy.

■ Quick Revision Cheat Sheet

Topic	Key Fact
OSI Layers	7: App/Pres/Sess/Trans/Net/DLL/Phy — "All People Seem To Need Data Processing"
TCP Port	Well-known: 0-1023 | Registered: 1024-49151 | Dynamic: 49152-65535
IPv4 classes	A: /8, B: /16, C: /24 | Private: 10/8, 172.16/12, 192.168/16
DNS default port	UDP 53 (queries), TCP 53 (zone transfers > 512 bytes)
Ethernet frame	Preamble+SFD(8) + Dest MAC(6) + Src MAC(6) + Type(2) + Data(46-1500) + FCS(4)
TTL	IPv4 header field; decremented by 1 at each router; 0 → packet dropped + ICMP sent
ARP	Layer 2.5; IP→MAC; Gratuitous ARP: host announces own IP/MAC (used for failover)
DHCP lease	DORA: Discover→Offer→Request→ACK | Renewal at 50% of lease time
BGP	Only EGP; used between Autonomous Systems; "glue of the internet"; TCP port 179
TLS 1.3	Removed: RSA key exchange, SHA-1, MD5, CBC ciphers | Added: 0-RTT, ECDHE mandatory

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